

Statistical Inference and Data Analysis (CIE 457) Course Project

MLE Estimation and Estimator Distribution in Linear Regression

1 Project Objective

The goal of this group project (2-3 members per group) is to develop a rigorous understanding of **Maximum Likelihood Estimation (MLE)** in linear regression, with emphasis on:

- The estimator as a **random variable**
- The **distribution** of the estimator
- The dependence of this distribution on **model assumptions**
- The role of **noise structure** (homoscedastic vs heteroscedastic)

Guiding Principle: The objective is not to fit a model, but to understand how the estimator behaves under different assumptions.

2 Model Framework

We consider the linear model:

$$y = X\beta + \epsilon$$

2.1 Homoscedastic Case

$$\epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

MLE estimate:

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

2.2 Heteroscedastic Case

$$\epsilon \sim \mathcal{N}(0, \Sigma), \quad \Sigma \neq \sigma^2 I$$

MLE estimate:

$$\hat{\beta}_{WLS} = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y$$

3 Project Tasks

3.1 MLE and Model Formulation (Theory)

- Write the likelihood function for:
 - Homoscedastic Gaussian noise
 - Heteroscedastic Gaussian noise
- Derive and explain the MLE estimator for:
 - Ordinary Least Squares (OLS)
 - Weighted Least Squares (WLS)
- State clearly:
 - All model assumptions
 - Why the estimator is a random variable
 - The distribution of $\hat{\beta}$ in both cases

3.2 Linear Regression under Homoscedastic Noise

- Assume:
$$\epsilon \sim \mathcal{N}(0, \sigma^2 I)$$
- Generate data:
$$y = X\beta + \epsilon$$
- Compute:
 - OLS estimator $\hat{\beta}$
- Study estimator distribution via repeated sampling from the true model (Monte Carlo):
 - Empirical mean
 - Empirical covariance

- Histogram/KDE of $\hat{\beta}$
- Compare with theoretical distribution:

$$\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(X^T X)^{-1})$$

3.3 Linear Regression under Heteroscedastic Noise

- Assume:

$$\epsilon \sim \mathcal{N}(0, \Sigma), \quad \Sigma \neq \sigma^2 I$$

- Generate data:

$$y = X\beta + \epsilon$$

- Compute:
 - OLS estimator
 - WLS estimator (MLE under correct model)
- Study estimators (OLS and WLS) distributions:
 - Empirical mean
 - Empirical covariance
 - Distribution shape
- Compare:
 - OLS vs WLS performance
 - Effect of misspecified noise model
 - Efficiency differences

3.4 Inference from Estimator Distribution

- Construct confidence intervals for $\hat{\beta}$
- Construct prediction intervals
- Compare the two intervals in terms of what they mean and their application

3.5 Real Data Application

- Select a dataset (e.g., wage vs age, income vs experience)
- Fit linear or polynomial regression models

- Analyze:
 - Coefficients and variability
 - Residual structure
 - Evidence of heteroscedasticity

4 Code Requirements

- Implement OLS and WLS
- Implement simulation for estimator distribution
- Clearly map equations to code

5 Deliverables

- Presentation (15–20 minutes)
- Report (5–8 pages)
- Code

6 Grading (20 Points) — Task-Based Oral Evaluation

Important: Grading is performed through one-to-one discussion. Each task is evaluated independently during oral examination. Code and report are required for reference but are not graded directly; they are used to support questioning.

Task 1: Likelihood and Model Formulation (4 points)

- Interpret the likelihood function for the linear model
- State all assumptions (Gaussianity, covariance structure)
- Explain how OLS arises from MLE
- Explain how WLS arises from MLE under heteroscedasticity

Evaluation:

- 4: Complete derivation understanding and correct interpretation under questioning

- 3: Minor gaps in reasoning
- 2: Partial understanding or formula-based answers
- 1: Weak conceptual understanding
- 0: Cannot explain

Task 2: Estimator as a Random Variable (4 points)

You must explain:

- Why $\hat{\beta}$ is random
- Relationship between repeated sampling and estimator variability

Evaluation:

- 4: Clear probabilistic interpretation and strong reasoning
- 3: Good understanding with minor imprecision
- 2: Partial or inconsistent explanation
- 1: Deterministic view of estimator
- 0: No understanding

Task 3: Estimator Distribution and Simulation (5 points)

This task focuses on empirical characterization of the estimator distribution and comparison to theory.

- Explain simulation procedure
- Interpret empirical mean and covariance of $\hat{\beta}$
- Compare empirical results with theory
- Discuss distribution shape (Gaussianity, deviations)

Evaluation:

- 5: Strong empirical + theoretical match, full interpretation
- 4: Good understanding with minor gaps

- 3: Basic correct results, limited interpretation
- 2: Weak or incomplete analysis
- 1: Poor or unclear results
- 0: Missing or incorrect

Task 4: Homoscedastic vs Heteroscedastic Analysis (4 points)

- Analyze OLS under homoscedastic noise
- Compare OLS vs WLS under heteroscedastic noise
- Explain the effect of covariance structure on estimator efficiency

Evaluation:

- 4: Deep understanding of model mismatch and efficiency
- 3: Good understanding with minor gaps
- 2: Basic qualitative understanding only
- 1: Confused reasoning
- 0: No understanding

Task 5: Code and Implementation Mapping (2 points)

Students must:

- Map mathematical expressions to code implementation
- Explain simulation design choices

Evaluation:

- 2: Fully understands and can modify code reasoning
- 1: Partial understanding
- 0: Cannot explain code

Task 6: Experimental Insight and Interpretation (1 point)

Students must:

- Interpret simulation and real-data results coherently
- Connect observations back to theory

Evaluation:

- 1: Clear and consistent interpretation
- 0: Weak or unclear reasoning

Bonus Task: Fisher Information, Sufficient Statistics and Optimality (Max +2 Points)

This bonus is reserved for students who demonstrate a deep mathematical dive into the theoretical limits of estimation. To earn these points, you need to understand the following topics with proper presentation:

- **Cramér-Rao Lower Bound (CRLB):** Derive the Fisher Information Matrix $I(\beta)$ for the Gaussian linear model and show that $\text{Var}(\hat{\beta})$ achieves the CRLB.
- **Sufficiency:** Use the Factorization Theorem to identify the sufficient statistics for β and σ^2 , explaining their significance in data compression.

Evaluation:

- 2: Rigorous derivation and clear conceptual link to "estimator efficiency."
- 1: Correct conceptual explanation of the theorems without full derivation.
- 0: No attempt or fundamental misunderstanding.